The University of Lahore
Department of Computer Science & IT
Mid Term Examination

Spring-2024

Course Title:	Computer Organization & Assembly Language	Course Code:	CS11207	Credit Hours:	**4**
Course Instructors:	Mr. M. Yasin, Ms. Asmara, Ms. Fatima, Mr. Fiaz, Ms. Zarbakht	Program Name:	**BSCS**		
Time Allowed:	**60 Minutes**	**Maximum Marks:**	**20**		
Date:	**19-03-24**	Course Mentor:	**Mr. M. Yasin Nasir**		
Student's Name:		Signature:			
Reg. No:		Section:	**A,B,C,D,E,F**		

Important Instructions / Guidelines:

- **Remember! SOMEONE is always with you (Be Relaxed), and HE is also watching you (Be Honest)**
- Question Paper is **SELF EXPLANATORY**. Understanding the Question Paper is part of Solution Solve subjective part on the answer sheet provided separately.
- Your paper will be marked as **ZERO**, if involved in any kind of cheating.

Question # 1: (6 marks)

Write down the status Flags (1/0) of the following flags after execution of each piece of code initially all flags are Zero.

INSTRUCTIONS	DESCRIPTION	CF	PF	OF	ZF	SF	AF
OR AH, BL	**Where AX=A01Dh, BX=8C18h**						
NEG BH	**Where BX = 7F00h**						
SAR AX, CL	**Where AX = CF23h,CX=8904**						
ADD AX, DX	**Where AX =FFF h, DX = 1h**						
CMP AL, BH	**Where AX=$FFF0h$, BX = FFFF**						

Question # 2: (4 marks)

Consider the following Memory Area and give the answers of the following questions:-

0C21:01A0	**13 96 D0 E0 D0 E0 A2 1E - 99 80 3E 20 99 00 75 24**	**..........> ..u$**
0C21:01B0	**A2 24 99 0A C9 75 1D 0A - C0 74 19 8B 0E 21 96 E3**	**.$...u...t...!..**
0C21:01C0	**13 B0 1A 06 33 FF 8E 06 - 00 96 F2 AE 07 75 05 4F**	**....3........u.O**
0C21:01D0	**89 3E 21 96 BB 06 97 80 - 3E 13 96 00 74 03 BB 4C**	**.>!.....>...t..L**
0C21:01E0	**97 BE C3 96 8B 3E 05 98 - B9 08 00 AC 3C 3F 75 02**	**.....>......<?u.**
0C21:01F0	**8A 07 3C 20 74 01 AA 43 - E2 F1 B1 03 B0 20 38 04**	**..< t..C..... 8.**
0C21:0200	**74 12 B0 2E AA AC 3C 3F-75 02 8A 07 3C 20 74 01**	**t.....<?u...< t.**
0C21:0210	**AA 43 E2 F1 32 C0 AA C3-F6 46 04 02 75 43 8B A3**	**.C..2....F..uC..**

a.) Which Logical address (i.e segment and offset address) in the above Memory area is equal to Physical address 0C3EA?
b.) Find the physical address for the 0C21:01FF.

Question # 3: Write down assembly language program for the following: (6 marks)

<u>Input</u>: A string which is assumed to be initialized in data segment terminated by "$".
"5 Quick Brown Fox Jumps Over The 3 Lazy Dogs ☺ " (Note : ☺ = ":)")

<u>Action</u>:

- **You don't need to declare other string.**
- **Read the given string using any addressing mode. Add the numeric values which are being used in the string.**
- **Perform the case conversion (capital to small and vice versa) for each of the ALPHABET in the above string and replace the original ALPHABET with the converted ALPHABET in the memory.**
- **Replace the numeric values and ☺ with the space.**

<u>Display</u>: Display the modified string and numeric value at the output screen.

Question # 4: Find the output of the given assembly language (4 Marks)

Consider the Data area of a Program given below, how this data will be stored in the hexadecimal form (address/contents) in Memory? (Assume that VA begins at offset 0010)

VA DB 0ABh, 0CDh, 0EFh, 40h, 50h, 60h
VB DB 123D,1011B
VC DW 1234h, 0ABCDh
VD DD 45768923h
VE DW 5,2 Dup("A")
VF DB "CC" **(Note : ASCII code of character C is 43h)**

00 10	…	…	…	…	…	…	…	…	…	…	…	…	…	00 1E	00 1F
AB	CD	EF	40	50	60	7B	0B	34	12	CD	AB	23	89	76	45
00 20	00 21	…	…	…	…	…	…	…	…	…	…	…	00 2D	00 2E	00 2F
05	00	41	41	43	43										
00 30	00 31	00 32	…	…	…	…	…	…	…	…	…	…	…	…	00 3F
00 40	00 41	…	…	…	…	…	…	…	…	…	…	…	…	…	00 4F